

Trust Region Methods

In different algorithms such as conservative policy iteration and policy gradient methods, the updates are performed incrementally.

* Similarly, the trust region methods aim to optimize the objective function while keeping the updates incremental.

$$\begin{aligned} \theta_{t+1} &= \arg \min_{\theta} / \arg \max_{\theta} \tilde{J}_{\theta_t}(\theta) && \text{approximate objective at } \theta_t \\ \text{s.t. } &D(\theta_t, \theta) \leq \delta && \text{trust-region constraint} \end{aligned}$$

Kullback-Leibler (KL) Divergence: The KL-divergence $D_{KL}(P \parallel Q)$ of probability measures P and Q , defined over the same sample space $\mathcal{X}$ measures how different P is from Q .

For discrete distributions, KL-divergence is defined as

$$\begin{aligned} D_{KL}(P \parallel Q) &= \mathbb{E}_{x \sim P} \left[\log \frac{P(x)}{Q(x)} \right] = -\mathbb{E}_{x \sim P} \left[\log \frac{Q(x)}{P(x)} \right] \\ &= \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)} = \sum_{x \in \mathcal{X}} P(x) \underbrace{(\log P(x) - \log Q(x))}_{\text{difference}} \end{aligned}$$

KL-divergence satisfies the following properties:

- $D_{KL}(P \parallel Q) \geq 0$
- $P = Q \iff D_{KL}(P \parallel Q) = 0 = D_{KL}(Q \parallel P)$

Idea: Knowing θ_t at time t , we compute θ_{t+1} based on

$$\begin{aligned} \max_{\theta} \quad & \mathbb{E}_{s \sim \bar{p}_{\mu_0}}^{\pi_{\theta_t}} \left[\mathbb{E}_{a \sim \pi_{\theta}} \left[A^{\pi_{\theta_t}}(s, a) \right] \right] && \text{maximizing improvement over current policy (performance difference lemma)} \\ \text{subject to} \quad & \underbrace{D_{KL}(\theta_t \parallel \theta)}_{\text{shorthand notation}} \leq \delta && \text{trust-region constraint} \end{aligned}$$

where the constraint ensures that the new policy does not diverge from the current policy much in terms of the distribution over trajectories.

We write (define) the KL-divergence in the constraint as:

$$D_{KL}(\theta_t \parallel \theta) := \underbrace{\mathbb{E}_{s \sim \bar{p}_{\mu_0}}^{\pi_{\theta_t}}}_{\text{averaged over states}} \left[D_{KL}(\pi_{\theta_t} \parallel \pi_{\theta}) \right]$$

$$\begin{aligned} \text{definition of } D_{KL} &\leftarrow \mathbb{E}_{s \sim \bar{p}_{\mu_0}}^{\pi_{\theta_t}} \left[\mathbb{E}_{a \sim \pi_{\theta_t}(\cdot|s)} \left[\log \frac{\pi_{\theta_t}(a|s)}{\pi_{\theta}(a|s)} \right] \right] \\ &= \mathbb{E}_{s, a \sim \bar{\pi}_{\mu_0}^{\pi_{\theta_t}}} \left[\log \frac{\pi_{\theta_t}(a|s)}{\pi_{\theta}(a|s)} \right] \end{aligned}$$

Natural Policy Gradient: The original TRPO formulation is difficult to solve. Instead of that, we form an approximation of the optimization problem (the objective and the constraint), which is easier to solve.

* Consider the objective function

$$f(\theta) = \mathbb{E}_{s \sim \bar{\rho}_{\theta_t}}^{\pi_{\theta_t}} \left[\mathbb{E}_{a \sim \pi_{\theta}}^{\pi_{\theta_t}} [A^{\pi_{\theta_t}}(s, a)] \right]$$

and its first-order Taylor approximation around θ_t

$$\tilde{f}(\theta) = f(\theta_t) + \underbrace{(\nabla_{\theta} f(\theta_t))^T}_{\text{gradient}} (\theta - \theta_t) .$$

One can show that

- $f(\theta_t) = 0$

- $\nabla_{\theta} f(\theta_t) = \mathbb{E}_{s, a \sim \bar{\rho}_{\theta_t}}^{\pi_{\theta_t}} \left[\underbrace{\nabla_{\theta} \log \pi_{\theta_t}(a|s)}_{\text{score}} A^{\pi_{\theta_t}}(s, a) \right]$

$$\longrightarrow \tilde{f}(\theta) = \left(\mathbb{E}_{s, a \sim \bar{\rho}_{\theta_t}}^{\pi_{\theta_t}} \left[\nabla_{\theta} \log \pi_{\theta_t}(a|s) A^{\pi_{\theta_t}}(s, a) \right] \right)^T (\theta - \theta_t)$$

Proof:

$$f(\theta) = \mathbb{E}_{s \sim \bar{\rho}_{\mathcal{M}_0}^{\pi_{\theta_t}}} \left[\mathbb{E}_{a \sim \pi_{\theta}} \left[A^{\pi_{\theta_t}}(s, a) \right] \right]$$

- $f(\theta_t) = \mathbb{E}_{s \sim \bar{\rho}_{\mathcal{M}_0}^{\pi_{\theta_t}}} \left[\mathbb{E}_{a \sim \pi_{\theta_t}} \left[A^{\pi_{\theta_t}}(s, a) \right] \right] = 0$

- $\nabla_{\theta} f(\theta_t) = \nabla_{\theta} \left[\mathbb{E}_{s \sim \bar{\rho}_{\mathcal{M}_0}^{\pi_{\theta_t}}} \left[\mathbb{E}_{a \sim \pi_{\theta}} \left[A^{\pi_{\theta_t}}(s, a) \right] \right] \right]_{\theta=\theta_t}$

$$= \mathbb{E}_{s \sim \bar{\rho}_{\mathcal{M}_0}^{\pi_{\theta_t}}} \left[\nabla_{\theta} \left[\mathbb{E}_{a \sim \pi_{\theta}} \left[A^{\pi_{\theta_t}}(s, a) \right] \right]_{\theta=\theta_t} \right]$$

$$= \mathbb{E}_{s \sim \bar{\rho}_{\mathcal{M}_0}^{\pi_{\theta_t}}} \left[\nabla_{\theta} \left[\sum_{a \in \mathcal{A}} \pi_{\theta}(a|s) A^{\pi_{\theta_t}}(s, a) \right]_{\theta=\theta_t} \right]$$

log-derivative ←
trick

$$= \mathbb{E}_{s \sim \bar{\rho}_{\mathcal{M}_0}^{\pi_{\theta_t}}} \left[\sum_{a \in \mathcal{A}} \nabla_{\theta} \log \pi_{\theta}(a|s) \Big|_{\theta=\theta_t} \pi_{\theta_t}(a|s) A^{\pi_{\theta_t}}(s, a) \right]$$

$$= \mathbb{E}_{s \sim \bar{\rho}_{\mathcal{M}_0}^{\pi_{\theta_t}}} \left[\mathbb{E}_{a \sim \pi_{\theta_t}} \left[\nabla_{\theta} \log \pi_{\theta_t}(a|s) A^{\pi_{\theta_t}}(s, a) \right] \right]$$

* Consider the trust-region constraint

$$g(\theta) = D_{KL}(\theta_t \parallel \theta)$$

and its second-order Taylor approximation around θ_t

$$\tilde{g}(\theta) = g(\theta_t) + \underbrace{(\nabla_{\theta} g(\theta_t))^T}_{\text{gradient}} (\theta - \theta_t) + \frac{1}{2} (\theta - \theta_t)^T \underbrace{\nabla_{\theta}^2 g(\theta_t)}_{\text{Hessian}} (\theta - \theta_t) .$$

One can show that

- $g(\theta_t) = 0_{1 \times 1}$

- $\nabla_{\theta} g(\theta_t) = 0_{d \times 1}$

Fisher Information Matrix

- $\nabla_{\theta}^2 g(\theta_t) = \mathbb{E}_{s, a \sim \bar{\pi}_{\mu_0}^{\theta_t}} \left[\underbrace{(\nabla_{\theta} \log \pi_{\theta_t}(a|s))}_{\text{score}} (\nabla_{\theta} \log \pi_{\theta_t}(a|s))^T \right]$

$$\rightarrow \tilde{g}(\theta) = \frac{1}{2} (\theta - \theta_t)^T \mathbb{E}_{s, a \sim \bar{\pi}_{\mu_0}^{\theta_t}} \left[(\nabla_{\theta} \log \pi_{\theta_t}(a|s)) (\nabla_{\theta} \log \pi_{\theta_t}(a|s))^T \right] (\theta - \theta_t)$$

Proof:

$$g(\theta) = D_{KL}(\theta_t \parallel \theta)$$

- $g(\theta_t) = D_{KL}(\theta_t \parallel \theta_t) = 0$

- $$\begin{aligned}\nabla_{\theta} g(\theta_t) &= \nabla_{\theta} \left[\mathbb{E}_{s, a \sim \bar{\nu}_{\mu_0}^{\pi_{\theta_t}}} \left[\log \frac{\pi_{\theta_t}(a|s)}{\pi_{\theta}(a|s)} \right] \right]_{\theta=\theta_t} \\&= \mathbb{E}_{s, a \sim \bar{\nu}_{\mu_0}^{\pi_{\theta_t}}} \left[\nabla_{\theta} \left[\log \pi_{\theta_t}(a|s) - \log \pi_{\theta}(a|s) \right]_{\theta=\theta_t} \right] \\&= \mathbb{E}_{s, a \sim \bar{\nu}_{\mu_0}^{\pi_{\theta_t}}} \left[- \nabla_{\theta} \log \pi_{\theta}(a|s) \Big|_{\theta=\theta_t} \right] \\&= - \mathbb{E}_{s \sim \bar{\rho}_{\mu_0}^{\pi_{\theta_t}}} \left[\sum_{a \in \mathcal{A}} \pi_{\theta_t}(a|s) \nabla_{\theta} \log \pi_{\theta_t}(a|s) \right]\end{aligned}$$

derivative of $\log(\cdot)$ $\leftarrow = - \mathbb{E}_{s \sim \bar{\rho}_{\mu_0}^{\pi_{\theta_t}}} \left[\sum_{a \in \mathcal{A}} \nabla_{\theta} \pi_{\theta_t}(a|s) \right]$

Linearity of gradient $\leftarrow = - \mathbb{E}_{s \sim \bar{\rho}_{\mu_0}^{\pi_{\theta_t}}} \left[\nabla_{\theta} \left[\sum_{a \in \mathcal{A}} \pi_{\theta_t}(a|s) \right]_{\theta=\theta_t} \right]$

$$= - \mathbb{E}_{s \sim \bar{\rho}_{\mu_0}^{\pi_{\theta_t}}} \left[\nabla_{\theta} [1]_{\theta=\theta_t} \right] = 0$$

- $\nabla_{\theta}^2 g(\theta_t) = \nabla_{\theta}^2 \left[\mathbb{E}_{s,a \sim \bar{r}_{\mathcal{M}_0}^{\pi_{\theta_t}}} \left[\log \frac{\pi_{\theta_t}(a|s)}{\pi_{\theta}(a|s)} \right] \right]_{\theta=\theta_t}$

$$= \mathbb{E}_{s,a \sim \bar{r}_{\mathcal{M}_0}^{\pi_{\theta_t}}} \left[- \nabla_{\theta}^2 \log \pi_{\theta}(a|s) \Big|_{\theta=\theta_t} \right]$$

$$\begin{aligned} &= - \mathbb{E}_{s \sim \bar{p}_{\mathcal{M}_0}} \left[\sum_{a \in \mathcal{A}} \pi_{\theta_t}(a|s) \frac{\nabla_{\theta}^2 \pi_{\theta_t}(a|s) \pi_{\theta_t}(a|s) - \nabla_{\theta} \pi_{\theta_t}(a|s) (\nabla_{\theta} \pi_{\theta_t}(a|s))^T}{\pi_{\theta_t}^2(a|s)} \right] \end{aligned}$$

$$\nabla_{\theta} \log \pi_{\theta} = \frac{\nabla_{\theta} \pi_{\theta}}{\pi_{\theta}}$$

$$\nabla_{\theta}^2 \log \pi_{\theta} = \frac{(\nabla_{\theta}^2 \pi_{\theta}) \pi_{\theta} - (\nabla_{\theta} \pi_{\theta})(\nabla_{\theta} \pi_{\theta})^T}{\pi_{\theta}^2}$$

$$= - \mathbb{E}_{s \sim \bar{p}_{\mathcal{M}_0}} \left[\sum_{a \in \mathcal{A}} \nabla_{\theta}^2 \pi_{\theta_t}(a|s) \right]$$

$$+ \mathbb{E}_{s,a \sim \bar{r}_{\mathcal{M}_0}^{\pi_{\theta_t}}} \left[\frac{\nabla_{\theta} \pi_{\theta_t}(a|s)}{\pi_{\theta_t}(a|s)} \left(\frac{\nabla_{\theta} \pi_{\theta_t}(a|s)}{\pi_{\theta_t}(a|s)} \right)^T \right]$$

Linearity of
gradient
+
derivative of
 $\log(\cdot)$

$$\leftarrow \mathbb{E}_{s,a \sim \bar{r}_{\mathcal{M}_0}^{\pi_{\theta_t}}} \left[(\nabla_{\theta} \log \pi_{\theta_t}(a|s)) (\nabla_{\theta} \log \pi_{\theta_t}(a|s))^T \right]$$

* Therefore, with the first-order approximation of the objective function and the second-order approximation of the constraint, the original optimization problem

$$\begin{aligned} \max_{\theta} \quad & \mathbb{E}_{s \sim \bar{\rho}_{\theta_0}}^{\pi_{\theta_t}} \left[\mathbb{E}_{a \sim \pi_{\theta}} \left[A^{\pi_{\theta_t}}(s, a) \right] \right] \\ \text{subject to} \quad & D_{KL}(\theta_t \| \theta) \leq \delta \end{aligned}$$

can be converted to

$$\begin{aligned} \max_{\theta} \quad & \overbrace{(\nabla_{\theta} J(\pi_{\theta_t}))^T}^{\text{gradient of RL objective}} (\theta - \theta_t) \\ \text{subject to} \quad & (\theta - \theta_t)^T \underbrace{F_{\theta_t}}_{\text{Fisher Information matrix}} (\theta - \theta_t) \leq \delta. \end{aligned}$$

In this new optimization problem,

- the objective function is linear and
- the constraint is quadratic and convex.

→ The closed-form solution can be obtained by

natural gradient:
 $F_{\theta_t}^{-1} \nabla_{\theta} J(\pi_{\theta_t})$

$$\theta_{t+1} = \theta_t + \underbrace{\eta}_{\text{step size / learning rate}} \underbrace{F_{\theta_t}^{-1}}_{\text{(Moore-Penrose) pseudoinverse}} \nabla_{\theta} J(\pi_{\theta_t})$$

Pre-conditioned gradient ascent update

$$\text{where, } \eta = \sqrt{\frac{\delta}{(\nabla_{\theta} J(\pi_{\theta_t}))^T F_{\theta_t}^{-1} (\nabla_{\theta} J(\pi_{\theta_t}))}}.$$

Natural Policy Gradient Algorithm

- Initialize θ_0

- For $i = 0, 1, 2, \dots$

similar to other policy gradient methods

- Estimate the policy gradient $\nabla_{\theta} J(\pi_{\theta_i})$ by g_i

- Estimate the Fisher information matrix F_{θ_i} by

$$F_i = (\nabla_{\theta} \log \pi_{\theta_i}(a|s)) (\nabla_{\theta} \log \pi_{\theta_i}(a|s))^T$$

- Update θ

$$\theta_{i+1} = \theta_i + \eta F_i^{-1} g_i$$

Proximal Policy Optimization: Implementing the TRPO algorithm is computationally expensive. The PPO algorithm aims to reduce computational complexity by using a surrogate objective and a clipping operation.

Recall the TRPO Formulation:

$$\begin{aligned} \max_{\theta} \quad & \mathbb{E}_{s \sim \bar{\rho}_{\mu_0}^{\pi_{\theta_t}}} \left[\mathbb{E}_{a \sim \pi_{\theta}} \left[A^{\pi_{\theta_t}(s, a)} \right] \right] \\ \text{subject to} \quad & D_{KL}(\theta_t \| \theta) \leq \delta \end{aligned}$$

- It would be helpful to rewrite the objective function as

$$\mathbb{E}_{s \sim \bar{\rho}_{\mu_0}^{\pi_{\theta_t}}} \left[\mathbb{E}_{a \sim \pi_{\theta_t}} \left[\underbrace{\frac{\pi_{\theta}(a|s)}{\pi_{\theta_t}(a|s)}}_{\text{importance sampling weight}} A^{\pi_{\theta_t}(s, a)} \right] \right]$$

because it enables using samples from the current policy π_{θ_t} for sampling-based estimation of the objective.

- By treating the trust-region constraint as a soft constraint, the optimization problem becomes

$$\max_{\theta} \left[\mathbb{E}_{s \sim \bar{\rho}_{\mu_0}}^{\pi_{\theta_t}} \left[\mathbb{E}_{a \sim \pi_{\theta_t}} \left[\frac{\pi_{\theta}(a|s)}{\pi_{\theta_t}(a|s)} A^{\pi_{\theta_t}}(s, a) \right] \right] \right. \\ \left. - \lambda D_{KL}(\theta_t \| \theta) \right]$$

constraint turned into penalty

One version of the PPO algorithm uses stochastic gradient ascent to approximately solve this optimization. In this case, it adjusts the penalty coefficient λ adaptively.

- To further simplify the process, another version of the PPO algorithm applies a clipping operation to imitate the trust-region constraint. In this case, the optimization problem becomes

$$\max_{\theta} \mathbb{E}_{s \sim \bar{\rho}_{\mu_0}}^{\pi_{\theta_t}} \left[\mathbb{E}_{a \sim \pi_{\theta_t}} \left[\min \left(\frac{\pi_{\theta}(a|s)}{\pi_{\theta_t}(a|s)} A^{\pi_{\theta_t}}(s, a), \right. \right. \right. \\ \left. \left. \left. \text{clip} \left(\frac{\pi_{\theta}(a|s)}{\pi_{\theta_t}(a|s)}, 1-\epsilon, 1+\epsilon \right) A^{\pi_{\theta_t}}(s, a) \right) \right] \right]$$

min of two terms

penalty for large policy update $\epsilon \geq 0$

maximize a lower bound on the original objective

$$\text{clip}(x, L, u) = \begin{cases} L & \text{if } x < L \\ x & \text{if } L \leq x \leq u \\ u & \text{if } x > u \end{cases}$$